Candidate name:

_____

Date: _____

# Unofficial NUFYPET Math Simulation

Time: 1 hour (60 min.) Version 9.1
Total Marks: ____/30

Instructions

- Circle the answer you find right
- Calculators **MUST NOT** be used
- Figures are **NOT** accurately drawn and **NOT** to scale, except it is mentioned in a question
- There is only one answer for each question, so answers with two or more selected options marked as zero
- If you think something is wrong with the test, report a Methodist

Information

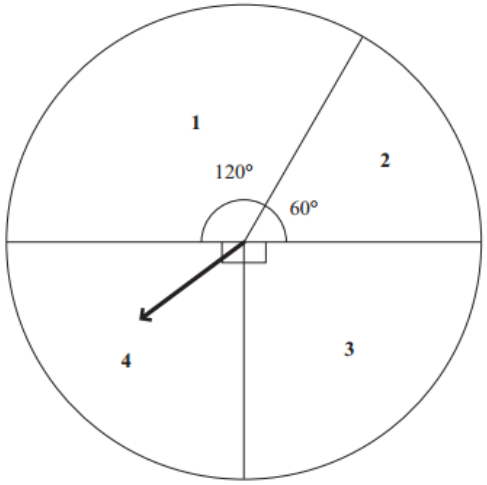
- The total mark for test is 30 (Each question marked as 1 point)
- There is **NO** penalty for wrong answers

Advice

- Read each question carefully
- Follow the time (There is an average 2 minutes for each question)
- Don't stay in one task for long
- Try to answer for all questions
- Recheck your answers if possible

Note: This is unofficial test made by MSDC

**1**. The figure shows a fair, circular spinner with 4 sectors numbered 1, 2, 3 and 4. The angle of sector 1 is 120°, the angle of sector 2 is 60° and the angle of each of sectors 3 and 4 is 90°.The spinner is to be spun. The number of the sector in which the spinner's pointer stops is called the score.



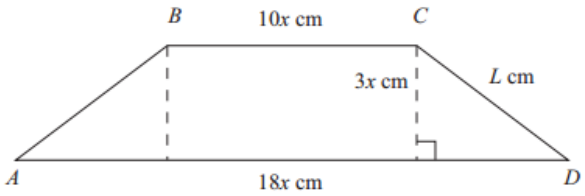

Calculate the probability that the score is 1 or 3.

**A** 7/12 **B** 7/24 **C** 7/48 **D** 7/20 **E** 7/17

**2**. Which one of the following is a factor of $6x^3 + 7x^2 - 18x + 5$ ?

**A** $x+1$ **B** $x+2$ **C** $x+3$ **D** $x-1$ **E** $x-2$

**3.** The figure shows a symmetrical trapezium with $AD = 18x\ cm$, $BC = 10x\ cm$ and $BA = CD = L\ cm$. The perpendicular distance between $AD$ and $BC$ is $3x\ cm$.




Find an expression, in terms of $x$, for the area of $ABCD$.

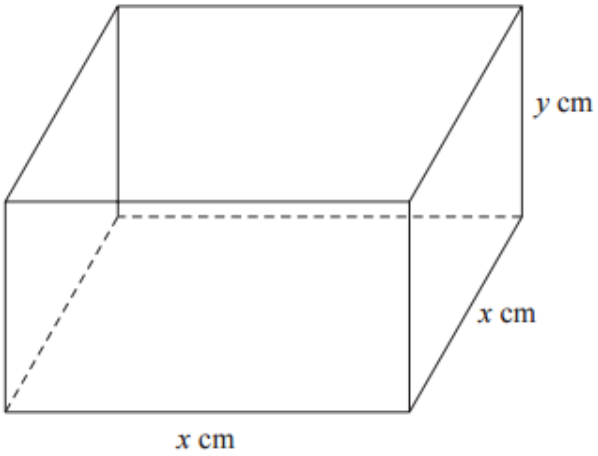
**A** $168x^2$ **B** $10.5x^2$ **C** $21x^2$ **D** $42x^2$ **E** $84x^2$

**4.** All of 60 different vitamin pills contain at least one of the vitamins $A, B$ and $C$. Twelve have $A$ only, 7 have $B$ only, and 11 have $C$ only. Six have all three vitamins.

If there are $x$ pills having $A$ and $B$ only; $x$ pills having $B$ and $C$ only; and $x$ pills having $A$ and $C$ only, then how many pills contain vitamin $A$?

**A** 34 **B** 32 **C** 30 **D** 36 **E** 38

**5.** The figure below shows a metal box with no top.




The four sides and the base of the box are to be cut from a single rectangular sheet of metal of width $x\ cm$ as shown in the second figure below.

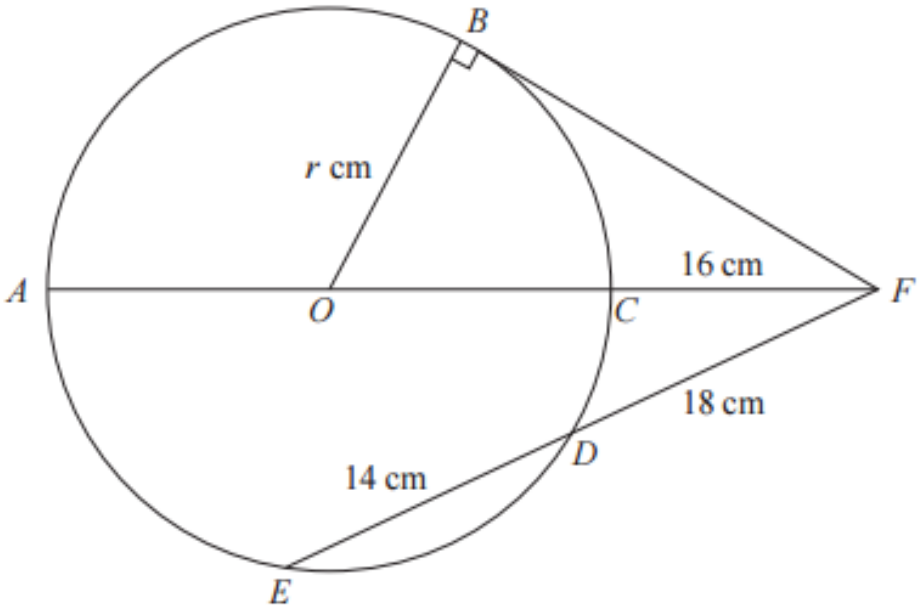



The area of the sheet of metal is $S\ cm^2$ and the volume of the metal box is $40\ cm^3$.

Assuming that no metal is wasted when the box is made, which one is the correct expression for $S$ in terms of $x$?

**A** $S = x^2 + \frac{160}{x}$ **B** $S = x^2 + \frac{x}{160}$ **C** $S = x^2 + \frac{160}{x^2}$ **D** $S = x + \frac{160}{x^2}$ **E** $S = x^2 + \frac{320}{x}$

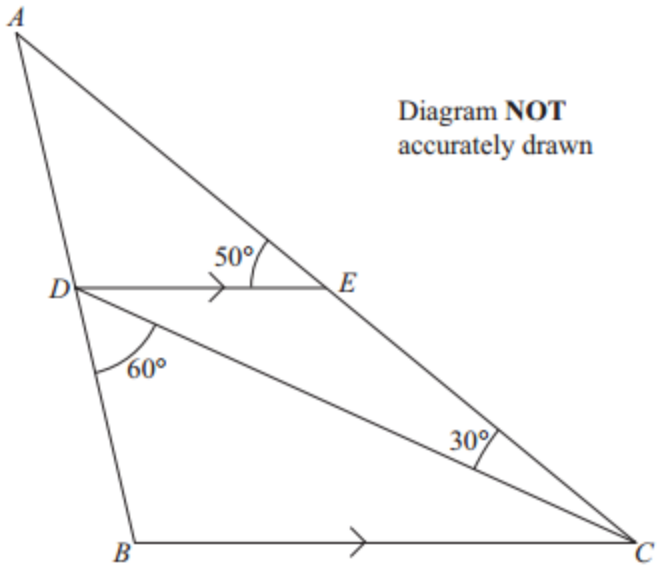
**6.** $ABCDE$ is a circle centre $O$. The diameter, $AC$, is extended to the point $F$ so that $CF = 16\,cm$. The line $BF$ is the tangent to the circle at $B$ and $FDE$ is a straight line such that $FD = 18\,cm$ and $DE = 14\,cm$. The radius of the circle is $r\,cm$.




What is the value of $r$?

**A** $5\,cm$ **B** $10\,cm$ **C** $20\,cm$ **D** $2.5\,cm$ **E** $24\,cm$

**7.** In the figure, $D$ is a point on side $AB$ and $E$ is a point on side $AC$ of $\Delta ABC$ such that $DE$ and $BC$ are parallel. $\angle AED = 50°, \angle DCE = 30°$ and $\angle BDC = 60°$.




Find, in degrees, the size of $\angle BCD$.

**A** $10°$ **B** $20°$ **C** $30°$ **D** $40°$ **E** $50°$

**8.** The letters of the word "INDEPENDENT" are written on individual cards and the cards are put into a box. A card is selected and then replaced and then a second card is selected.

What is the probability of obtaining the letter "E" twice?

**A** 1/121 **B** 3/121 **C** 18/121 **D** 9/121 **E** 27/121

**9.** What of the following is equivalent to the above expression?

$$\frac{4\sqrt{12}+12\sqrt{108}-10\sqrt{128}-8\sqrt{50}}{10}$$

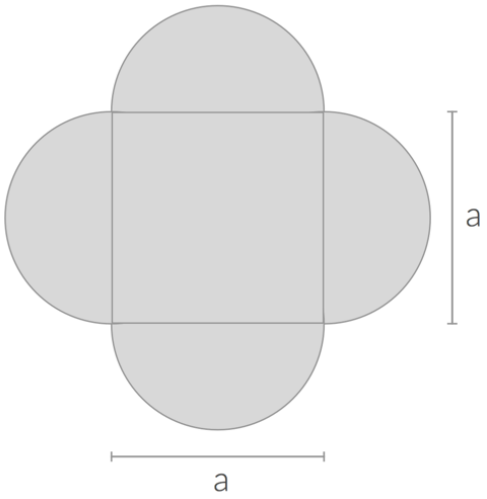
**A** $80\sqrt{3}+40\sqrt{2}$ **B** $40\sqrt{3}-80\sqrt{2}$ **C** $4\sqrt{2}-8\sqrt{3}$ **D** $8\sqrt{3}-4\sqrt{2}$ **E** $8\sqrt{3}-12\sqrt{2}$

**10.** Which of the following is equivalent to the above expression?

$$\frac{3\sqrt{2}}{\sqrt{3}-\sqrt{2}}$$

**A** $3\sqrt{6}+6$ **B** $\sqrt{6}-6$ **C** $\sqrt{6}+3$ **D** $3\sqrt{6}-6$ **E** $\sqrt{6}+6$

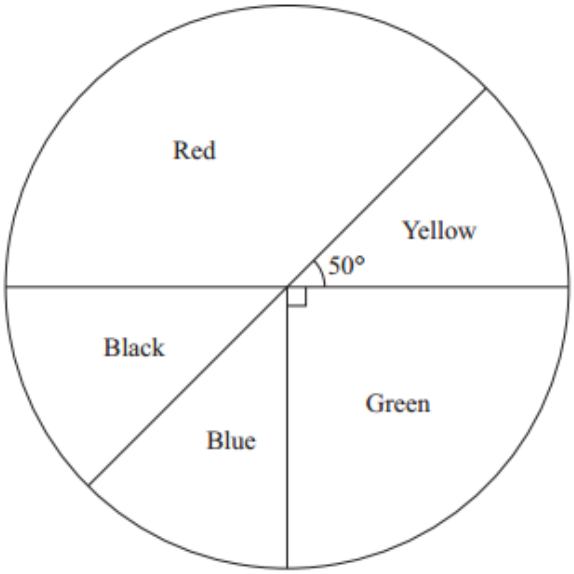
**11.** Consider the shape below.




Which expression best represents the shaded area?

**A** $a^2+\frac{1}{4}\pi a^2$ **B** $\frac{5}{4}\pi a^2$ **C** $a^2+\frac{1}{2}\pi a^2$ **D** $\frac{3}{2}\pi a^2$ **E** $3\pi a^2$

**12.** A company has 180 cars to sell. The colours of the cars are red, yellow, green, blue and black. The pie chart shows information about the number of cars of each colour. The angle of the sector for yellow cars is 50° and the angle for green cars is 90°. There are as many yellow cars as there are black cars and the total number of black and blue cars is the same as the number of green cars.




Calculate the number of black cars.

**A** 25 **B** 20 **C** 30 **D** 40 **E** 35

**13.** A bag contains 6 red marbles and 4 yellow marbles. A marble is drawn at random and not replaced. Two further draws are made, again without replacement.

What is the probability of drawing at least one red marble?

**A** 1/30 **B** 29/30 **C** 1/6 **D** 28/31 **E** 30/31

**14.** Consider two graphs with the following equations:

$$y = x^2 + bx + 4$$

$$y = 3x + 4$$

If the two graphs only have one point of intersection, what is the correct value for $b$?

**A** $4$ **B** $-3$ **C** $-1$ **D** $0$ **E** $3$

**15.** In a class of 50 students, 15 students are tremendously lazy and have chosen not to do any sports sports. The rest chose either running, swimming, or both.

In total, 25 students chose running and 25 students chose swimming.

If a student is selected randomly from the class, what is the probability that he/she chose both running and swimming?

**A** 1/5 **B** 1/3 **C** 2/3 **D** 3/10 **E** ½

**16.** The symbol ♦ defines a mathematical binary operation such that $a \blacklozenge b = \sqrt{ab}$ for all positive integers.

Evaluate the value of $(8 \blacklozenge 2) \blacklozenge (4 \blacklozenge 9)$.

**A** $2\sqrt{3}$ **B** $4\sqrt{3}$ **C** $2\sqrt{6}$ **D** $4\sqrt{6}$ **E** $\sqrt{6}$

**17.** From a group of 10 boys and 7 girls, 2 boys and 3 girls will be chosen randomly.

How many different combinations of boys and girls are possible?

**A** 1557 **B** 1575 **C** 1595 **D** 5175 **E** 5715

**18.** Imagine a solid cube with sides of length $3cm$.

This cube is magically cut in half. What is the surface area of each half?

**A** 9 $cm^2$ **B** 18 $cm^2$ **C** 27 $cm^2$ **D** 36 $cm^2$ **E** 54 $cm^2$

**19.** The volume of an imaginary cube is $216\ cm^3$.

What is the perimeter of this cube?

**A** 6 cm **B** 16 cm **C** 18 cm **D** 36 cm **E** 72 cm

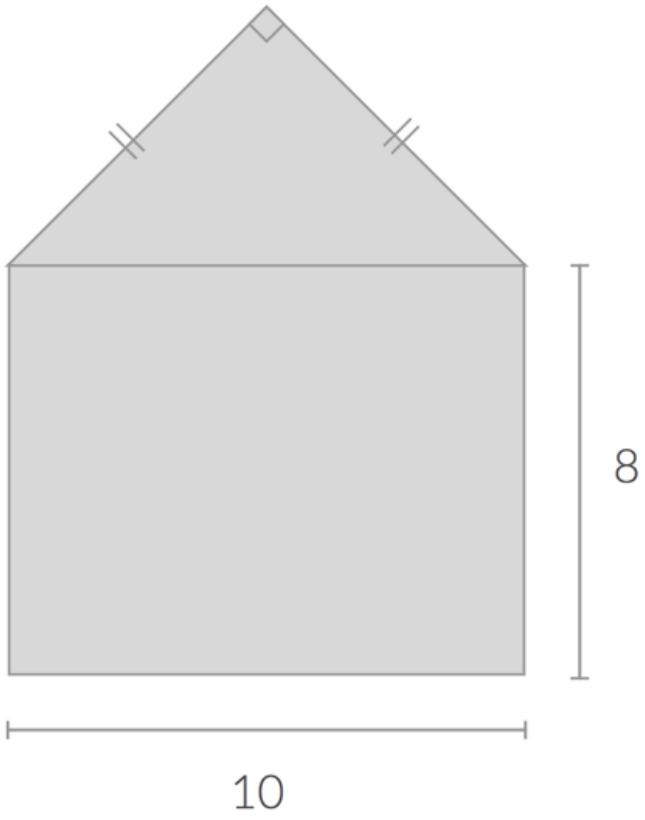
**20.** Ten years ago, Ron's age was twice that of Hermione.

Today, the ratio of their ages is 7:4.

What is the sum of their ages today?

**A** 90 **B** 100 **C** 110 **D** 120 **E** 125

**21.** The diagram below shows a school playing field.




What is the perimeter of the above field?

**A** $26 + 10\sqrt{2}$ **B** $26 + 5\sqrt{2}$ **C** $26 + \sqrt{2}$ **D** $36 + 10\sqrt{2}$ **E** $36 + 5\sqrt{2}$

**22.** Five years ago, Mother's age was 3 times that of Alex. 5 years in the future, Mother's age will be twice that of Alex.

How old is Mother today?

**A** 30 **B** 35 **C** 40 **D** 45 **E** 50

**23.** Mrs. Robinson's math class has an average exam score of 70. The average of students who scored below 60 is 50, and the average of students who scored above 60 is 75. There are 20 students in the class.

How many students scored below 60?

**A** 4 **B** 6 **C** 8 **D** 16 **E** 20

**24.** Rearrange to make $x$ the subject of the equation.

$$y = \frac{4x - 3}{2x + 1}$$

**A** $x = \frac{y-2}{4-3y}$ **B** $x = \frac{y+2}{4-2y}$ **C** $x = \frac{y-3}{4-2y}$ **D** $x = \frac{y+3}{4-2y}$ **E** $x = \frac{y+3}{4+2y}$

**25.** Imagine two boxes. The first box contains 2 red balls and 3 white balls. The second box contains 5 red balls and 3 white balls.

A ball is drawn from each box at random.

What is the probability that the ball from the first box is red, and the ball from the second box is white?

**A** 1/40 **B** 3/20 **C** 3/8 **D** 2/5 **E** 3/40

**26.** Five bells commence tolling together and toll at intervals of 3, 6, 9, 12, and 15 seconds respectively.

In 60 minutes, how many times do they toll together?

**A** 10 times **B** 20 times **C** 30 times **D** 40 times **E** 50 times

**27.** A solid sphere with diameter 10 is placed inside a regular cylinder with length 10 and cross-sectional diameter 10.

What is ratio of the volume between the cylinder and the sphere?

**A** 1 : 2 **B** 2 : 1 **C** 1 : 1 **D** 3 : 2 **E** 2 : 3

**28.** Let $y = 1 + \sqrt{\frac{x+3}{x}}$, for all positive integers.

Rearrange the formula to give $x$ in term of $y$.

**A** $x = \frac{3}{y(y-2)}$ **B** $x = \frac{3}{(y-1)^2}$ **C** $x = \frac{3}{(y-2)^2}$ **D** $x = \frac{y(y-2)}{3}$ **E** $x = \frac{(y-2)^2}{3}$

**29.** An electronic device beeps after every 40 seconds. Another device beeps after 42 seconds. They beep together at 10 am.

When will they next beep together?

**A** 10.14 am **B** 10.16 am **C** 10.20 am **D** 10.28 am **E** 10.42 am

**30.** The lowest common multiple of three different numbers is 2520.

Which of the following cannot be their highest common factor?

**A** 35 **B** 40 **C** 49 **D** 56 **E** 72

Answer key:

| № | Correct Answer | Your answer |
|---|---|---|
| 1 | A | |
| 2 | D | |
| 3 | D | |
| 4 | A | |
| 5 | A | |
| 6 | B | |
| 7 | B | |
| 8 | D | |
| 9 | E | |
| 10 | A | |
| 11 | C | |
| 12 | A | |
| 13 | B | |
| 14 | E | |
| 15 | D | |
| 16 | C | |
| 17 | B | |
| 18 | D | |
| 19 | E | |
| 20 | C | |
| 21 | A | |
| 22 | B | |
| 23 | A | |
| 24 | D | |
| 25 | B | |
| 26 | B | |
| 27 | D | |
| 28 | A | |
| 29 | A | |
| 30 | C | |
| | | Total: _____ |